# Temperature and particle motion

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use a particle model to distinguish average motion from total energy.

**SUCCESS CRITERIA - I can...**

1. Describe temperature in terms of microscopic motion.
2. Use kelvin in proportional temperature comparisons.
3. Avoid treating a single particle speed as the temperature.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Sunlight carries energy across 3 million km of nearly empty space in a simplified example. Identify the transfer mechanism and explain why matter is unnecessary.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $K_{avg} \propto T$; $T(K) = T(°C) + 273.15$.
Ideal gas: mean translational kinetic energy per particle scales with absolute temperature. ∝ means proportional to.

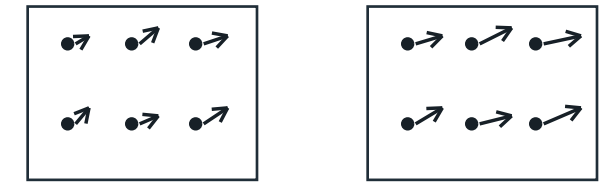
**Worked reasoning:** In a gas model, warmer temperature corresponds to greater average random molecular kinetic energy. Individual particles still have a range of speeds.

**Your turn:** Draw two equal-size samples using dots and arrows; label arrows as particle velocities and make the warmer sample's average arrow length greater.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

Sketch particles in samples at lower and higher temperature. Use arrow lengths to represent a range of speeds, with a larger average for the hotter sample.



A B

Each dot is a particle. Each arrow shows that particle's velocity.

**Marzano check - circle:** ① ② ③

## 04 Build the solution

An ideal gas warms from 300 to 600 K. By what factor does average translational kinetic energy per particle change?

**Marzano check - circle:** ① ② ③

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### 05 Use the idea

A student doubles 20°C to 40°C and claims average particle kinetic energy doubles. Explain why the scale matters.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

Two students perform the same lift. One takes 4 times as long as the other. Compare work and average power.

**Marzano check - circle:** 1 2 3

### 07 Exit check

At equal kelvin temperature, does a gas sample with twice as many particles have twice the average energy per particle?

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

Two gas samples have the same temperature but different volumes. Must the particle speeds differ? Explain what temperature alone fixes in this model.

**Marzano check - circle:** 1 2 3